

Semiconductor Devices and Circuits

Sections 20–21

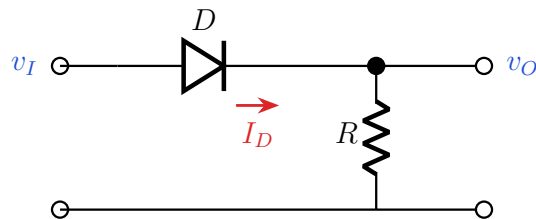
Problems for Sections 15–16 (batteries and resistor networks) are now in the Circuit Theory Foundations problem set.

Section 20 — Diodes

In every problem, validate your assumptions after solving.

Problem 20.1

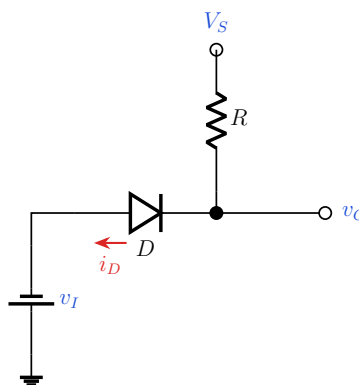
For the circuit below, if $v_I = 5\text{ V}$ and $V_{DO} = 0.7\text{ V}$, find v_O .



Problem 20.2

For the circuit below, find v_O in each of the following cases. Take $V_{DO} = 0.7\text{ V}$ throughout.

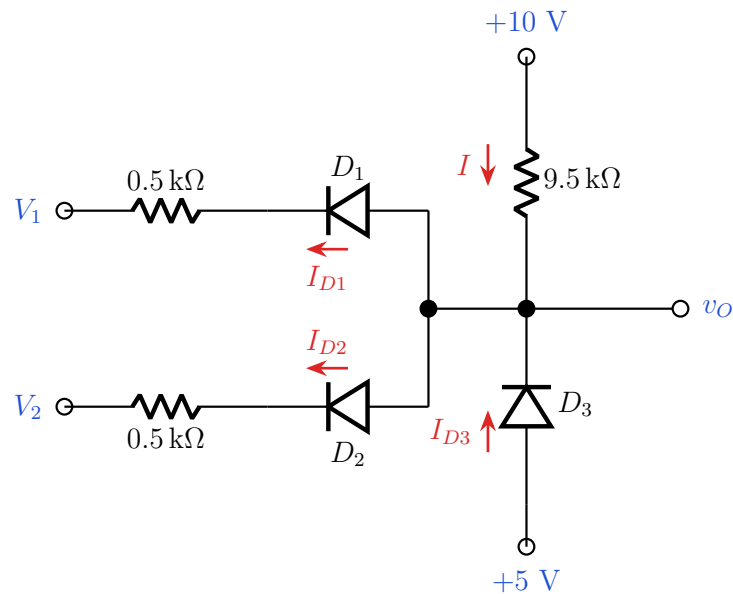
- (a) $v_I = 5\text{ V}$, $V_S = 10\text{ V}$
- (b) $v_I = 11\text{ V}$, $V_S = 10\text{ V}$
- (c) $v_I = 0\text{ V}$, $V_S = 10\text{ V}$



Problem 20.3

Analyse the circuit below to find I_{D1} , I_{D2} , I_{D3} and I for each of the following input combinations. Use the **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50$ Ω .

- (a) $V_1 = 0$ V, $V_2 = 2$ V
- (b) $V_1 = V_2 = 0$ V
- (c) $V_1 = V_2 = 5$ V
- (d) $V_1 = 5$ V, $V_2 = 0$ V

**Problem 20.4**

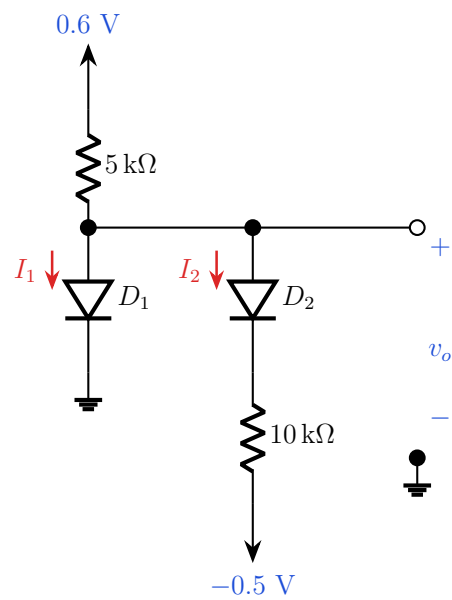
For the same circuit as Problem 20.3, replace the 9.5 k Ω pull-up resistor with a resistor of value X (in k Ω) and take $V_{DO} = 0.6$ V.

- (a) Repeat the analysis of Problem 20.3 for $X = 9.5$.
- (b) For what value of X will D_3 be in cut-off?

Problem 20.5

Analyse the circuit below to find I_1 , I_2 and v_o under each of the following diode models.

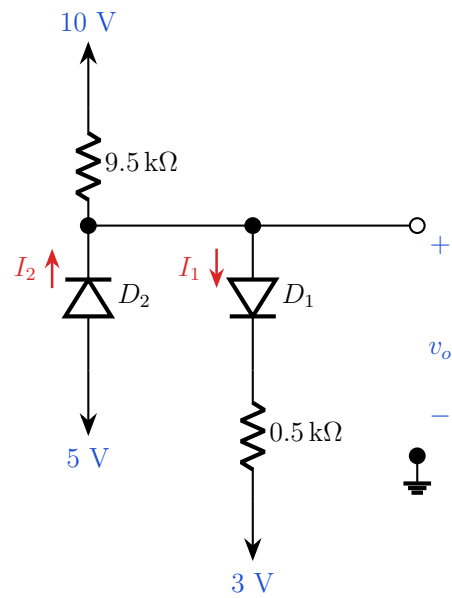
- (a) All diodes assumed **ideal**.
- (b) **CVD model** with $V_{D1} = 0.5$ V and $V_{D2} = 0.7$ V.
- (c) **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50$ Ω .



Problem 20.6

Analyse the circuit below to find I_1 , I_2 and v_o under each of the following diode models.

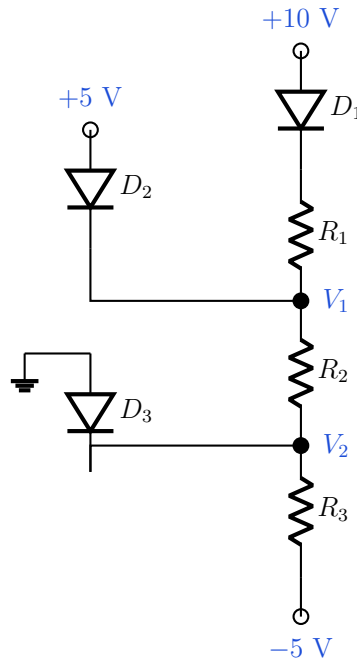
- (a) All diodes assumed **ideal**.
- (b) **CVD model** with $V_{DO} = 0.6$ V.
- (c) **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50$ Ω .



Problem 20.7

For the circuit below, take $V_{DO} = 0.6$ V.

- Determine R_1 , R_2 and R_3 such that $I_{D1} = 0.2$ mA, $I_{D2} = 0.3$ mA and $I_{D3} = 0.5$ mA.
- Find V_1 , V_2 and each diode current for $R_1 = 10$ k Ω , $R_2 = 4$ k Ω and $R_3 = 2.2$ k Ω .

**Problem 20.8**

For the same circuit as Problem 20.7 with $V_{DO} = 0.6$ V, find V_1 , V_2 and each diode current for:

- $R_1 = 3$ k Ω , $R_2 = 6$ k Ω , $R_3 = 2.5$ k Ω
- $R_1 = 6$ k Ω , $R_2 = 3$ k Ω , $R_3 = 6$ k Ω

Section 21 — Rectifiers

Validate your assumptions where a diode state must be chosen.

Problem 21.1

A voltage waveform $v_i = 10 \sin(100\pi t)$ V is input to a full-wave rectifier with a load resistance of $R = 50 \text{ k}\Omega$. Silicon diodes are used in this circuit, for which the forward drop is $V_{D_0} = 0.7 \text{ V}$.

- Show** the circuit of the rectifier. **Label** the input and output voltages properly. [2]
- Calculate** the DC value of the output voltage. [1]
- Contrast** the value found in part (b) with that when a $5 \mu\text{F}$ capacitor is connected in parallel with the load. [2]
- Identify** the two diodes that will be ON in the positive half cycle. [1]

Now the two diodes from part (d) are replaced with germanium diodes ($V_{D_0} = 0.2 \text{ V}$).

- Explain** the change in the voltage transfer characteristics and output voltage waveform of the circuit. Hence, calculate the peak of the output voltage in this case. [3+1]

Problem 21.2

A voltage waveform $v_i = 5 \sin(200\pi t)$ V is fed into a full-wave rectifier with a load resistor $R = 5 \text{ k}\Omega$. Silicon diodes are used in this circuit, where $V_{D_0} = 0.6 \text{ V}$.

- Draw** the rectifier circuit. **Label** the input and output voltages properly. Briefly **explain** the application of the circuit. [1+1+1]
- Calculate** the DC value of the output voltage, V_{dc} , and the output frequency, f_o . [1+1]
- Draw** the Voltage Transfer Characteristics (VTC) of the full-wave rectifier and **label** it properly. [2]
- Now you have to connect a capacitor in parallel with the load resistor. You have two capacitors of $5 \mu\text{F}$ and $1 \mu\text{F}$ at your disposal. Which capacitor will you use? **Explain** briefly with necessary calculations. [3]
- [Bonus]** A different input waveform is fed into the full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated in part (d) with the $5 \mu\text{F}$ capacitor. The new output frequency is 300 Hz. **Determine** the equation of the input waveform. [2]

Problem 21.3

A voltage waveform $v_i = 10 \sin(1000\pi t)$ V is fed into a full-wave (FW) rectifier with a load resistance $R = 10 \text{ k}\Omega$. A capacitor is also connected in parallel with the load to reduce the fluctuation of the output voltage. It produces a peak-to-peak ripple voltage which is **3%** of the peak output voltage. The diodes have a forward voltage drop of $V_{D_0} = 0.8 \text{ V}$.

- (a) **Deduce** the peak output voltage, V_p , and the peak-to-peak ripple voltage, $v_{r(p-p)}$. [2]
- (b) **Calculate** the average (DC) value of the output voltage. [1]
- (c) **Estimate** the value of the capacitor from the given data. [2]

Now the input voltage v_i is changed to the one shown in Figure (ii) of the original handout, and the **capacitor is removed**.

- (d) **Show** the output waveform and **indicate** the voltage levels properly. [2]

Problem 21.4

A voltage waveform $v_i = 8 \sin(2000\pi t)$ V is input to a full-wave rectifier. A resistance of $R = 50 \text{ k}\Omega$ is connected at the load. Assume that the diodes used in the circuit have a forward drop of 0.8 V .

- (a) **Draw** the circuit of the full-wave rectifier. Label the input and output voltages properly. [1]
- (b) **Draw** the waveforms of the input and output voltages. What are the peak values of input and output? Show them in the graph. [1+1]
- (c) **Find** the average voltage measured at the output. [1]